

Abdul Rafi

Computer Science & Engineering Undergraduate

Robotics & Autonomous Systems • Artificial Intelligence • Computer Vision • Embedded Systems

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University of Moratuwa, Sri Lanka

Profile

Computer Science and Engineering undergraduate at the University of Moratuwa with a strong interest in software engineering, distributed systems, full-stack development, and intelligent systems. I enjoy designing and building reliable software from requirements and system architecture through implementation, integration, and deployment. My experience includes full-stack web and mobile applications, backend APIs, databases, event-driven systems, and ROS 2-based robotics. I am particularly interested in developing scalable, maintainable software systems and combining software engineering with AI, computer vision, and robotics to solve real-world problems.

Education

University of Moratuwa, Sri Lanka March 2024 – Present

B.Sc. Engineering (Hons.) in Computer Science & Engineering

Specialization in Integrated Computer Engineering

CGPA: 3.15 / 4.00

St. Thomas' College, Matara

January 2009 – January 2023

G.C.E. Advanced Level (Physical Science Stream)

- Achieved 3 A grades (Combined Mathematics, Physics, Chemistry)
- All-Island Rank: 206 | Z-Score: 2.5139

Engineering Projects

School Management System

🐙 [GitHub](#)

Real Client Project | React | FastAPI | Supabase | PostgreSQL | Mobile + Push Notifications

- Built a full-stack attendance and student management platform (React + TypeScript, React Native, FastAPI, Supabase/PostgreSQL) for a school of approximately 700 students, used in production by real teachers and staff.
- Implemented role-based access, scheduled push notifications, and automated Excel/PDF reporting; owned the project end-to-end from requirements to deployment.

Kandypack Distribution System

🐙 [GitHub](#)

Semester 03 Group Project | React | Node.js/Express | MySQL

- Built a full-stack logistics management system for a hybrid rail-and-road supply chain, replacing manual Excel-based tracking with a relational database, REST API, and React dashboard.
- Designed a MySQL schema with triggers and stored procedures to enforce business rules: order lead-time validation, train capacity limits, staff working-hour caps, and schedule conflict prevention.
- Implemented JWT-based role-based access (Admin/Driver/Assistant), automated order-to-train-trip allocation, and reporting on sales, driver hours, and truck usage.

OnTime Distributed Transit System

🐙 [GitHub](#)

Semester 04 Group Project | Distributed Systems | Kafka | MQTT

- Developed Kafka-based event-driven communication for a distributed transit platform.
- Integrated and configured the Mosquitto MQTT broker for real-time messaging between services.
- Contributed to the system's distributed messaging and communication infrastructure.

Unitree G1 Humanoid Robot – Localization & Navigation

🐙 [GitHub](#)

Unitree G1 | ROS 2 | Humanoid Robotics | Localization | LLM + RAG

- Working with the Unitree G1 humanoid robot in simulation on robotic localization and navigation, integrating perception, sensing, and autonomous behaviors.
- Implemented ROS 2-based sensor integration and simulated robotic experimentation to validate autonomous navigation performance.
- Integrated an LLM-powered Retrieval-Augmented Generation (RAG) system to ground model responses in retrieved knowledge.

Technical Skills

Programming Languages – Python, Java, JavaScript, TypeScript, C++, C

Web Development – React, React Native, Node.js, FastAPI

Databases – MySQL, PostgreSQL, MongoDB

AI & Computer Vision (Working Knowledge) – OpenCV, LangChain, LLMs, RAG, YOLO

Embedded Systems & Hardware – STM32, Arduino, ESP32, FPGA (VHDL)

Robotics & Autonomous Systems – ROS 2, Nav2, SLAM, LiDAR, ArduPilot, MAVLink, Gazebo, MuJoCo

Engineering Tools – Git, Docker, CMake, Linux (Ubuntu)

Leadership & Service

- Programme Committee Member – Hit the Ground Tournament, University of Moratuwa
- Logistics Committee Member – SLIoT Competition, University of Moratuwa
- Logistics & Marketing Team Member – Career Fair 26
- Religious Coordinator – Majlis-Ul-Islam, University of Moratuwa

Achievements

- Runner-up – IEEE INSL Idea Stage 2025 Competition
- Top 50 – SPARK Competition – ROS 2 Autonomous Bathymetric Survey System with ArduPilot
- Finalist – YARL Geek Challenge Season 14
- Finalist – Sri Lanka Robotics Challenge (SLRC) 2025

References

Prof. Chandana D. Gamage

Professor

University of Moratuwa, Sri Lanka

B.Sc. Eng. (Moratuwa), M.Eng. (AIT),

Ph.D. (Monash), CEng, MIE(SL), MACM

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Dr. S. J. Sooriyaarachchi

Senior Lecturer

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